

Forecasting VN30 Index Volatility Using GARCH-Family Models: Evidence from Vietnam's Stock Market

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Abstract—This study examines whether asymmetric GARCH specifications improve the modeling and forecasting of daily VN30 index volatility relative to symmetric GARCH benchmarks. Using daily VN30 closing prices from January 2015 to April 2026, the price series is transformed into percentage log returns and split chronologically into training, validation, and test samples. The training sample is used to document stylized facts, including return stationarity, fat-tailed innovations, volatility clustering, and ARCH effects. A set of ARCH, GARCH, EGARCH, and GJR-GARCH models with normal and Student-t innovations is then estimated on the training sample. In-sample evidence favors the AR(1)-GJR-GARCH(1,1)-t model, suggesting persistent volatility, fat tails, and a statistically meaningful asymmetric response to negative shocks. However, validation-based forecast evaluation selects AR(1)-GARCH(1,1)-t as the final forecasting specification, with a QLIKE value only marginally lower than the simpler GARCH(1,1)-t model. On the test sample, the selected model outperforms historical variance and rolling 21-day variance benchmarks by QLIKE and most other forecast metrics. The findings show that asymmetry is important for explaining VN30 volatility dynamics, but it does not necessarily translate into superior one-step-ahead volatility forecasts in the validation period. The results support a cautious distinction between in-sample volatility interpretation and out-of-sample forecasting performance.

Index Terms—VN30 index, volatility forecasting, GARCH, GJR-GARCH, asymmetric volatility, Student-t innovations, QLIKE, Vietnam stock market.

I. INTRODUCTION

Volatility is a central object of interest in financial time-series analysis because it summarizes the uncertainty and risk associated with asset returns. For investors, portfolio managers, and risk monitors, an accurate understanding of volatility dynamics is often more operationally useful than a point forecast of the return itself. This is particularly relevant for emerging equity markets, where volatility may change rapidly across market regimes and where negative shocks can trigger stronger risk responses than positive shocks of the same magnitude.

This work was conducted as part of the TSA – Time Series Analysis.

The VN30 index represents a basket of large and liquid stocks in Vietnam and is widely used as a benchmark for the performance of the Vietnamese equity market. Like many equity indices, VN30 returns exhibit stylized facts that challenge constant-variance models: returns fluctuate around zero, extreme observations occur more often than under the normal distribution, and periods of large movements tend to cluster together. These features motivate the use of ARCH and GARCH-family models, which allow conditional variance to evolve over time as a function of past shocks and past volatility.

The research direction of this report is deliberately broader than simply applying GARCH to forecast volatility. The study asks whether adding an asymmetric volatility channel improves the explanation of VN30 volatility dynamics and whether this improvement in in-sample dynamics also improves out-of-sample forecasting performance. This distinction is important because a model that explains historical volatility clustering and leverage effects well may not necessarily be the model that produces the best one-step-ahead volatility forecasts.

A. Research Gap

The international literature provides extensive evidence that GARCH-family models are useful for volatility forecasting, including in Asian markets. However, results are often sensitive to the market, the loss function, the forecast design, and the data frequency. For Vietnam, existing studies more often examine spillovers, market connectedness, or broader volatility transmission rather than a focused train-validation-test comparison of symmetric versus asymmetric GARCH models for the VN30 index. This report addresses that narrower empirical gap by separating in-sample volatility explanation from validation- and test-based forecast evaluation.

B. Research Questions

This study is guided by four research questions:

- 1) Do VN30 daily returns exhibit stylized facts that justify GARCH-family volatility modeling, such as stationarity in returns, volatility clustering, fat tails, and ARCH effects?

- 2) Does an asymmetric GARCH specification, particularly GJR-GARCH, provide a better in-sample description of VN30 volatility dynamics than symmetric GARCH models?
- 3) Does the asymmetric specification that performs well in-sample also improve one-step-ahead volatility forecasting performance on a validation sample?
- 4) Does the final selected GARCH-family model outperform simple volatility benchmarks on a clean test sample?

C. Contributions

The report contributes in four ways. First, it provides a leakage-aware empirical design by splitting the sample chronologically into training, validation, and test periods. Second, it distinguishes between in-sample model adequacy, assessed by information criteria and residual diagnostics, and out-of-sample forecasting performance, assessed by validation and test losses. Third, it evaluates whether asymmetric volatility dynamics are useful for both explanation and forecasting, rather than assuming that these two objectives produce the same model choice. Finally, it offers practical evidence for VN30 risk monitoring by comparing the final GARCH forecast with historical and rolling-variance benchmarks.

II. LITERATURE REVIEW

A. ARCH and GARCH Foundations

Engle [1] introduced the ARCH model to represent time-varying conditional variance through lagged squared innovations. Bollerslev [2] generalized this idea by proposing the GARCH model, where conditional variance depends both on previous shocks and on previous conditional variance. This structure is especially suitable for financial returns because it captures volatility persistence and clustering using a parsimonious dynamic variance equation.

B. Asymmetric GARCH Models

A limitation of symmetric GARCH models is that they treat positive and negative shocks of the same magnitude as having identical effects on future volatility. In equity markets, negative returns are often associated with stronger increases in risk. Glosten, Jagannathan, and Runkle [3] address this issue through the GJR-GARCH model, which adds an indicator for negative shocks. In this report, the GJR term is used to test whether bad news has a larger effect on VN30 volatility than good news. EGARCH, introduced by Nelson [8], is also considered as a robustness candidate during the training-sample model comparison, but it is not emphasized in the final forecast stage when diagnostics and model-selection results do not support carrying it forward.

C. Volatility Forecasting Evaluation

Volatility is not directly observable at the daily frequency. Following the common approach in the volatility forecasting literature, squared returns are used as a noisy proxy for realized variance [4]. Forecast performance is evaluated using several loss functions rather than one criterion only, because different metrics penalize errors differently. QLIKE is used as the primary selection criterion because it is widely used in volatility forecast evaluation and is less sensitive to noisy volatility proxies than some squared-error losses [5].

D. Asian and Vietnamese Market Context

Sahiner [6] compares GARCH-family models across Asian financial markets and shows that model ranking can depend on data frequency, forecast method, and loss function. This supports the need for explicit out-of-sample validation rather than relying only on in-sample fit. Vietnamese market studies, including Dang, Nguyen, and Vo [7], emphasize the relevance of volatility dynamics and spillovers in Vietnam. This report complements that literature by focusing on VN30 univariate volatility modeling and by directly comparing explanatory asymmetry with forecast performance.

The review is intentionally concise. The report uses a small set of closely related sources to avoid diluting the methodological focus: ARCH/GARCH foundations, asymmetric GARCH, volatility forecast evaluation, Asian-market evidence, and Vietnam-market context.

III. DATA AND EXPLORATORY ANALYSIS

A. Data Source and Preprocessing

The empirical analysis uses daily VN30 closing prices from 5 January 2015 to 29 April 2026. The raw dataset contains 2,824 observations and seven columns. Data preprocessing converts Vietnamese column names into standardized English names, parses trading dates, cleans price and volume fields, sorts the series chronologically, removes duplicate trading dates, and retains strictly positive closing prices. No duplicate dates or missing closing values remain after cleaning.

The daily log return is constructed as

$$r_t = 100[\log(P_t) - \log(P_{t-1})] = 100 \log\left(\frac{P_t}{P_{t-1}}\right), \quad (1)$$

where P_t is the VN30 closing price at date t . The first observation is lost when computing log returns, leaving 2,823 daily return observations. The return is expressed in percentage points to improve numerical stability in GARCH estimation.

B. Chronological Train-Validation-Test Split

To avoid look-ahead bias, the processed return data are split chronologically rather than randomly. The training sample is used for preliminary diagnostics and in-sample

TABLE I: Data summary.

Metric	Value	Metric	Value
Raw rows	2,824.00	Mean return (%)	0.0431
Cleaned observations	2,824.00	Std. dev. return (%)	1.2302
Processed return observations	2,823.00	Minimum return (%)	-7.0522
Start date	2015-01-05	Maximum return (%)	6.6700
End date	2026-04-29	Skewness	-0.8186
Duplicated dates after cleaning	0.0000	Kurtosis	7.8676

Note. The processed return sample starts one trading day after the raw price sample because log returns require lagged prices.

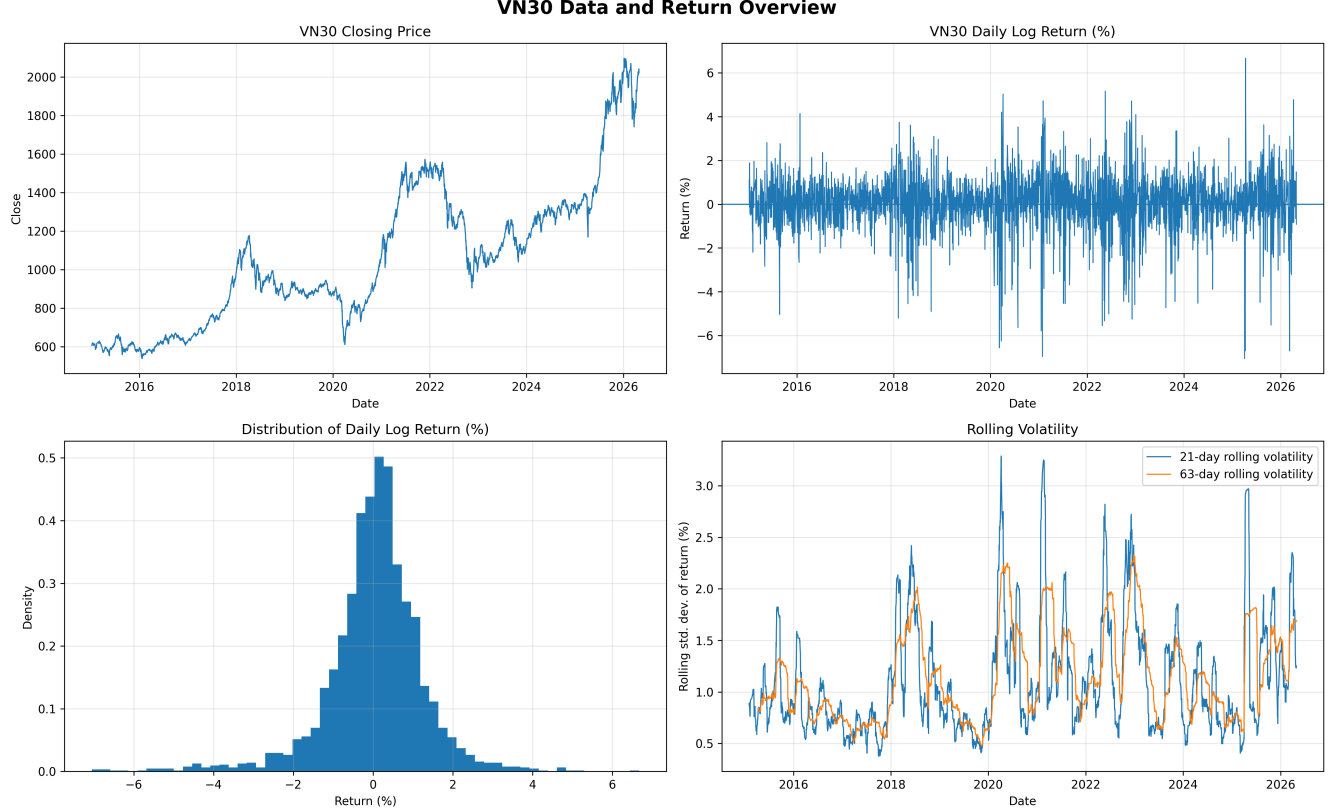


Fig. 1: VN30 price, return distribution, and rolling volatility overview.

model selection. The validation sample is used for forecast-based model selection. The test sample is reserved for final out-of-sample evaluation. No gap is inserted between adjacent samples because the target is one-step-ahead daily volatility; the first observation of the next split is a valid forecast target using information available up to the previous trading day.

C. Preliminary Diagnostics on the Training Sample

Formal diagnostics are conducted only on the training sample. The validation and test samples are treated as future data and are not used to justify model choice. The ADF test indicates that the VN30 closing price is non-stationary, while daily log returns are stationary. Ljung-Box tests show only mild return autocorrelation but very strong autocorrelation in squared returns. The ARCH-

LM test strongly rejects the null of no ARCH effects, confirming that conditional heteroskedasticity is present.

IV. METHODOLOGY

A. General Model Structure

Let r_t denote the daily VN30 log return in percent. The conditional mean is specified either as a constant mean or as an AR(1) mean. Innovations are decomposed into a conditional standard deviation and a standardized shock:

$$r_t = \mu + \phi r_{t-1} + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t. \quad (2)$$

The AR(1) term is included only to absorb weak linear dependence in the conditional mean. The main object of interest is the conditional variance σ_t^2 , which represents the model-implied one-step-ahead volatility state.

TABLE II: Chronological sample split.

Split	Period	Obs.	Mean return (%)	Std. return (%)	Min return (%)	Max return (%)	Gap
train	2015-01-06 to 2022-11-30	1976	0.0283	1.2218	-6.9624	5.1699	0
valid	2022-12-01 to 2024-08-13	423	0.0449	1.1794	-5.2557	4.7161	0
test	2024-08-14 to 2026-04-29	424	0.1100	1.3165	-7.0522	6.6700	0

Note. Gap equals zero trading observations because the forecast target is one-step-ahead daily volatility, not a multi-day forward realized-volatility window.

TABLE III: Preliminary tests on the training sample.

Test	Sample	Statistic	p-value
ADF test on VN30 close	Train only	-1.326	0.881
ADF test on VN30 return	Train only	-29.112	0.000
Ljung-Box on return, lag 20	Train only	34.049	0.026
Ljung-Box on squared return, lag 20	Train only	780.941	1.57e-152
ARCH-LM test, lag 10	Train only	263.149	9.28e-51

Note. Small p-values for squared returns and ARCH-LM provide direct motivation for GARCH-family volatility models.

VN30 Return Dependence and Volatility Diagnostics - Training Sample

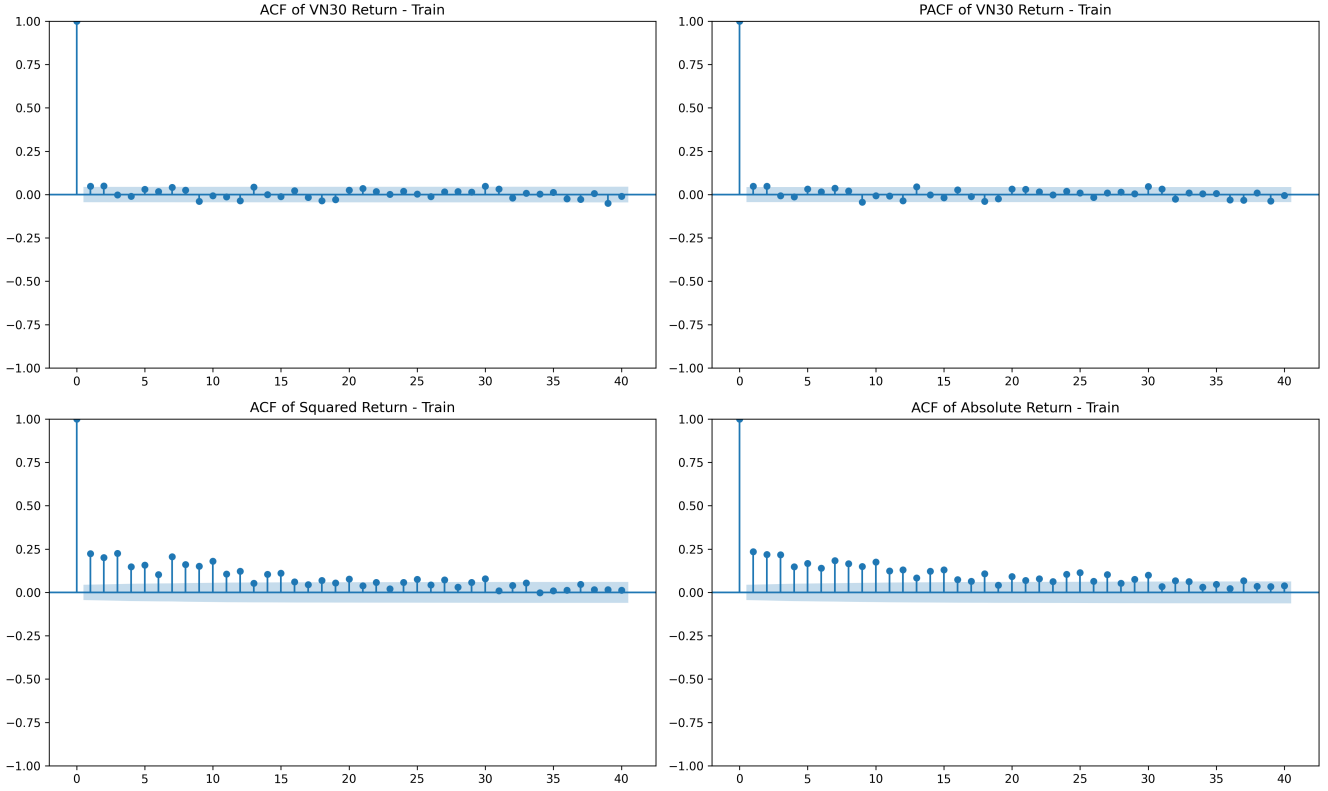


Fig. 2: Return dependence and volatility diagnostics on the training sample.

B. GARCH(1,1)

The symmetric GARCH(1,1) model captures volatility clustering by allowing conditional variance to depend on the previous squared innovation and the previous conditional variance:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (3)$$

Here α measures the short-run response of volatility to new shocks, while β measures volatility persistence. A high

$\alpha + \beta$ indicates that volatility shocks decay slowly.

C. GJR-GARCH(1,1)

The GJR-GARCH(1,1) model extends GARCH by allowing negative shocks to have an additional impact on future volatility:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma I(\varepsilon_{t-1} < 0) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (4)$$

If γ is positive and statistically significant, negative shocks raise future volatility more than positive shocks of the same magnitude. Under a symmetric shock distribution, the approximate persistence of a GJR-GARCH(1,1) process is $\alpha + \beta + \gamma/2$.

D. Student-t Innovations

Both normal and Student-t innovation distributions are considered in the training-sample model comparison. Student-t innovations are important because VN30 returns display fat tails and extreme observations. A smaller estimated degrees-of-freedom parameter indicates heavier tails. In the final candidate set, Student-t specifications are emphasized because they consistently provide better information criteria and align with the observed return distribution.

E. Candidate Model Specifications

F. Model Selection and Diagnostics

Training-sample selection uses a two-stage logic. First, candidate models are compared using log-likelihood, AIC, and BIC. BIC is emphasized because it penalizes model complexity more strongly. Second, standardized residuals are checked using Ljung-Box tests on residuals and squared residuals and an ARCH-LM test. A satisfactory volatility model should remove meaningful serial dependence from both standardized residuals and squared standardized residuals.

G. Forecasting Design

The forecasting task is one-step-ahead daily variance prediction. A model is first estimated on the training sample and used to recursively forecast the validation sample. At each validation date, the forecast is formed using information available up to the previous trading day. The realized return is then observed and used only to update the state for the next forecast. After the final GARCH model is selected by validation QLIKE, it is re-estimated on the combined training and validation sample and evaluated on the test sample.

Two benchmark forecasts are used. The historical variance benchmark uses the expanding variance of past returns, shifted by one day. The rolling benchmark uses the 21-day rolling variance of past returns, also shifted by one day. These shifts ensure that no contemporaneous return from the evaluation date enters the forecast.

H. Forecast Evaluation Metrics

Daily realized variance is proxied by squared returns:

$$RV_t = r_t^2. \quad (5)$$

For a variance forecast $h_t = \hat{\sigma}_t^2$, QLIKE is defined as

$$QLIKE = \frac{1}{n} \sum_{t=1}^n \left[\log(h_t) + \frac{RV_t}{h_t} \right]. \quad (6)$$

Lower QLIKE indicates better forecast performance. QLIKE is selected as the primary metric because it is specifically designed for volatility forecast evaluation and is robust to noise in the volatility proxy. MSE and MAE of variance forecasts are also reported, together with MAE of volatility forecasts based on square-rooted variances. The use of multiple metrics prevents the conclusion from depending entirely on one loss function.

V. EMPIRICAL RESULTS

A. Training-Sample Model Comparison

Table V reports the leading training-sample model comparison results. The AR(1)-GJR-GARCH(1,1)-t specification achieves the lowest BIC among all candidate models. The simpler GJR-GARCH(1,1)-t model is extremely close by BIC, indicating that the AR(1) mean adjustment improves the criterion only marginally. Nevertheless, the selected model is useful for interpreting volatility dynamics because it combines mean adjustment, asymmetric volatility, and fat-tailed innovations.

The selected training model implies high but mean-reverting volatility persistence. The estimated γ parameter is positive and statistically significant, supporting the presence of asymmetric volatility dynamics in VN30 returns. The estimated Student-t degrees of freedom is around 5.12, which is consistent with heavy-tailed return shocks. These findings answer the in-sample part of the research question: asymmetric GARCH helps explain VN30 volatility dynamics.

B. Residual Diagnostics and EGARCH Exclusion from Validation

The selected AR(1)-GJR-GARCH(1,1)-t model passes the main residual diagnostics on the training sample. Standardized residuals and squared standardized residuals no longer display significant serial dependence at the 5% level, and the post-model ARCH-LM test does not reject the absence of remaining ARCH effects.

EGARCH is not used in the validation-stage candidate set for two reasons. First, the EGARCH(1,1)-t specification has weaker training-sample model ranking than the leading GARCH and GJR-GARCH t models. Second, its squared-standardized-residual and ARCH-LM diagnostics remain significant at the 5% level, implying remaining volatility dependence. Although EGARCH with normal innovations passes these diagnostics, it is less consistent with the fat-tailed return evidence and has worse information criteria. Therefore, the validation exercise focuses on the diagnostically stronger Student-t GARCH and GJR-GARCH candidates.

C. Validation Forecast Selection

The validation stage answers a different question from the training stage. Instead of asking which model explains the training volatility dynamics best, it asks which candidate produces the most accurate one-step-ahead volatility

TABLE IV: Candidate model families.

Model family	Mean	Variance	Distribution	Purpose
ARCH(1), ARCH(5)	Constant mean	ARCH terms only	normal, Student-t	Baseline conditional heteroskedasticity models
GARCH(1,1), GARCH(1,2), GARCH(2,1)	Constant mean	Symmetric GARCH	normal, Student-t	Volatility persistence without asymmetry
AR(1)-GARCH(1,1)	AR(1) mean	Symmetric GARCH	normal, Student-t	Controls for weak mean dependence before modeling volatility
GJR-GARCH(1,1)	Constant mean	Asymmetric GARCH with negative-shock indicator	normal, Student-t	Tests whether bad news has a stronger volatility impact
AR(1)-GJR-GARCH(1,1)	AR(1) mean	Asymmetric GARCH	normal, Student-t	Main asymmetric candidate with mean adjustment
EGARCH(1,1)	Constant mean	Log-volatility asymmetric specification	normal, Student-t	Robustness candidate; not retained for validation when diagnostics are weaker

Note. All candidate models are estimated on the training sample. Validation is restricted to the diagnostically stronger Student-t GARCH and GJR-GARCH candidates.

TABLE V: Leading training-sample model comparison results.

Model	AIC	BIC	α	β	γ	GJR persistence	ν	AR(1)	Selected
AR(1)-GJR-GARCH(1,1)-t	5,711.27	5,750.39	0.041	0.804	0.169	0.929	5.125	0.052	Yes
GJR-GARCH(1,1)-t	5,717.34	5,750.87	0.047	0.811	0.149	0.933	5.025		No
GARCH(1,1)-t	5,734.81	5,762.75	0.130	0.838			4.917		No
AR(1)-GARCH(1,1)-t	5,731.27	5,764.80	0.132	0.835			4.969	0.033	No
EGARCH(1,1)-t	5,741.17	5,769.11	0.270	0.961			4.903		No
GARCH(1,2)-t	5,736.28	5,769.81	0.147	0.817			4.930		No
GARCH(2,1)-t	5,736.81	5,770.34	0.130	0.838			4.916		No
ARCH(5)-t	5,768.61	5,813.32	0.687				4.323		No

Note. Models are sorted by BIC. The full output is retained in the project tables. The close BIC values between AR(1)-GJR-GARCH(1,1)-t and GJR-GARCH(1,1)-t should be interpreted cautiously.

TABLE VI: Main parameter estimates for AR(1)-GJR-GARCH(1,1)-t.

Parameter	Estimate	p-value	Interpretation
ω	0.0887	0.0185	Long-run variance intercept
α_1	0.0408	0.0473	Immediate response to new shocks
β_1	0.8037	7.59e-53	Volatility persistence from the previous period
γ_1	0.1691	0.0084	Additional effect of negative shocks
ν	5.1246	1.42e-16	Degrees of freedom of Student-t innovations
AR(1)	0.0515	0.0490	Lagged return coefficient in the mean equation
$\alpha + \beta + \gamma/2$	0.9290		Approximate GJR volatility persistence

Note. The approximate GJR persistence is $\alpha + \beta + \gamma/2$. The positive γ term indicates that negative shocks have an additional volatility effect.

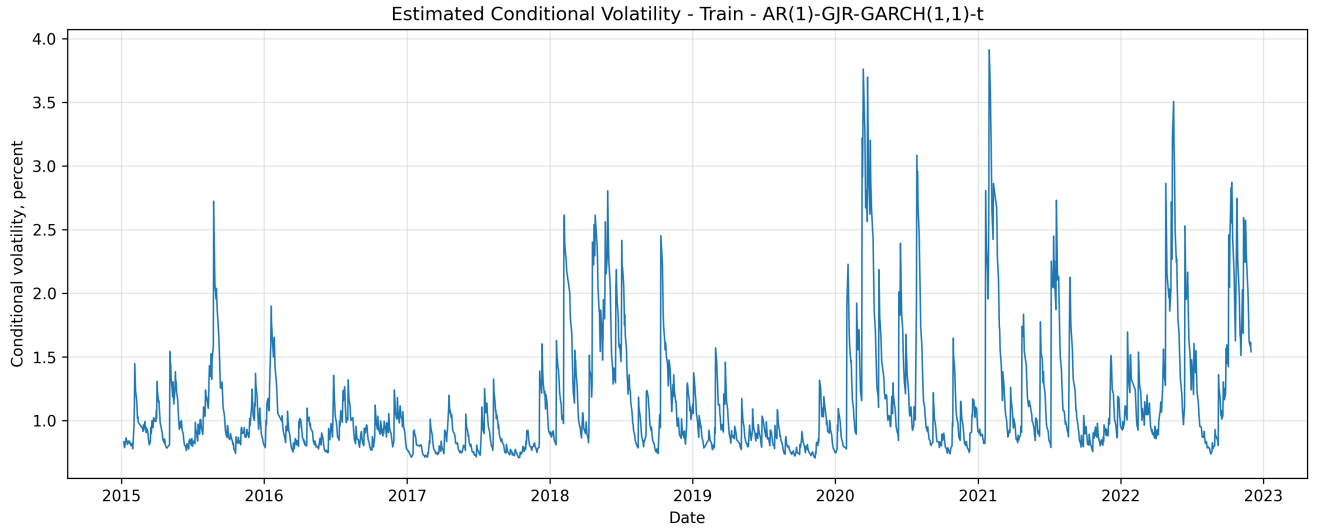


Fig. 3: Estimated conditional volatility from the selected training model.

TABLE VII: Residual diagnostics for selected and relevant candidate models.

Model	LB resid p	LB squared p	ARCH-LM p	Pass resid	Pass squared	Pass ARCH
AR(1)-GARCH(1,1)-t	0.6786	0.4373	0.4648	Yes	Yes	Yes
AR(1)-GJR-GARCH(1,1)-t	0.8728	0.4696	0.4670	Yes	Yes	Yes
EGARCH(1,1)-t	0.3390	0.0287	0.0362	Yes	No	No
GARCH(1,1)-t	0.3594	0.4051	0.4299	Yes	Yes	Yes
GJR-GARCH(1,1)-t	0.4573	0.4707	0.4740	Yes	Yes	Yes

Note. EGARCH(1,1)-t is shown because it was estimated during training but not carried forward to validation. It fails the squared-residual Ljung-Box and ARCH-LM diagnostics at the 5% level, while the GARCH and GJR-GARCH t-innovation candidates pass.

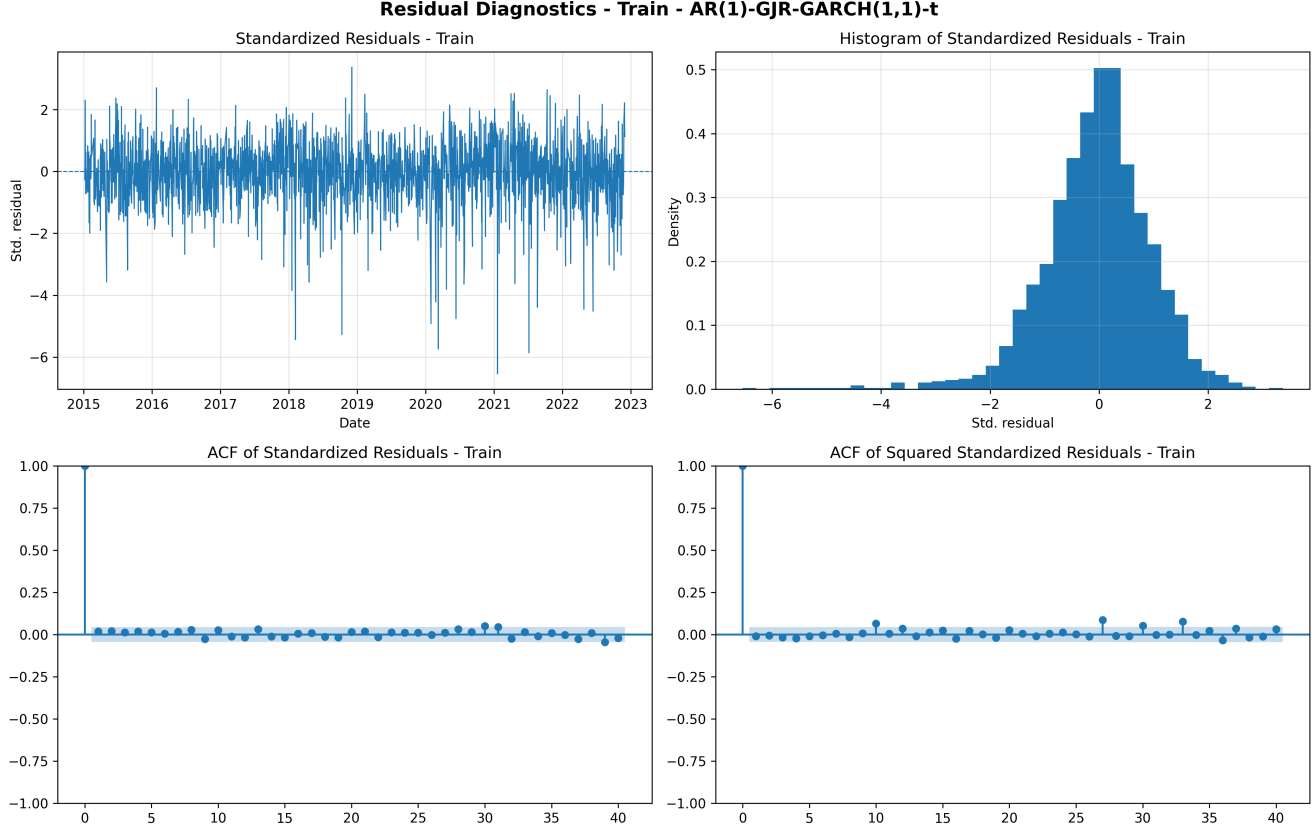


Fig. 4: Residual diagnostics for AR(1)-GJR-GARCH(1,1)-t on the training sample.

forecasts on future data. Table VIII shows that AR(1)-GARCH(1,1)-t achieves the lowest validation QLIKE and is selected as the final forecasting model.

The validation results show that the best in-sample explanatory model is not the best out-of-sample forecasting model. AR(1)-GJR-GARCH(1,1)-t is preferred for explaining asymmetric volatility dynamics on the training sample, but AR(1)-GARCH(1,1)-t has slightly better validation QLIKE. The difference between AR(1)-GARCH(1,1)-t and GARCH(1,1)-t is only 0.00004 in QLIKE, so the result should be interpreted as a near-tie among the symmetric t-innovation GARCH models rather than a large forecasting advantage. The asymmetric GJR candidates perform worse by validation QLIKE in this sample.

D. Test-Sample Evaluation

After validation selection, AR(1)-GARCH(1,1)-t is re-estimated on the combined training and validation sample and evaluated on the test sample. The test sample is used only once and is not involved in selecting the final model. Table IX reports that the selected GARCH model has the lowest QLIKE, MSE variance, and MAE volatility among the tested alternatives. Historical variance has a marginally lower MAE variance, but the difference is small and QLIKE strongly favors the GARCH specification.

The test plot illustrates a typical feature of volatility forecasting. The realized volatility proxy, $|r_t|$, is noisy and contains sudden spikes. The GARCH forecast is smoother because it estimates expected conditional volatility rather than the exact magnitude of tomorrow's return. After

TABLE VIII: Validation forecast performance.

Model	Obs.	MSE var.	MAE var.	QLIKE	MAE vol.	Selected
AR(1)-GARCH(1,1)-t	423	10.2466	1.6434	1.2151	0.7069	Yes
GARCH(1,1)-t	423	10.2151	1.6411	1.2152	0.7065	No
GJR-GARCH(1,1)-t	423	10.7306	1.6456	1.2468	0.7054	No
AR(1)-GJR-GARCH(1,1)-t	423	10.8641	1.6559	1.2548	0.7075	No
Rolling 21-day variance	423	10.0160	1.6593	1.2796	0.7016	No
Historical variance	423	10.4108	1.6726	1.3323	0.7541	No

Note. QLIKE is the primary selection criterion. The difference between AR(1)-GARCH(1,1)-t and GARCH(1,1)-t is extremely small, so the validation result should not be interpreted as overwhelming evidence for the AR(1) mean term.

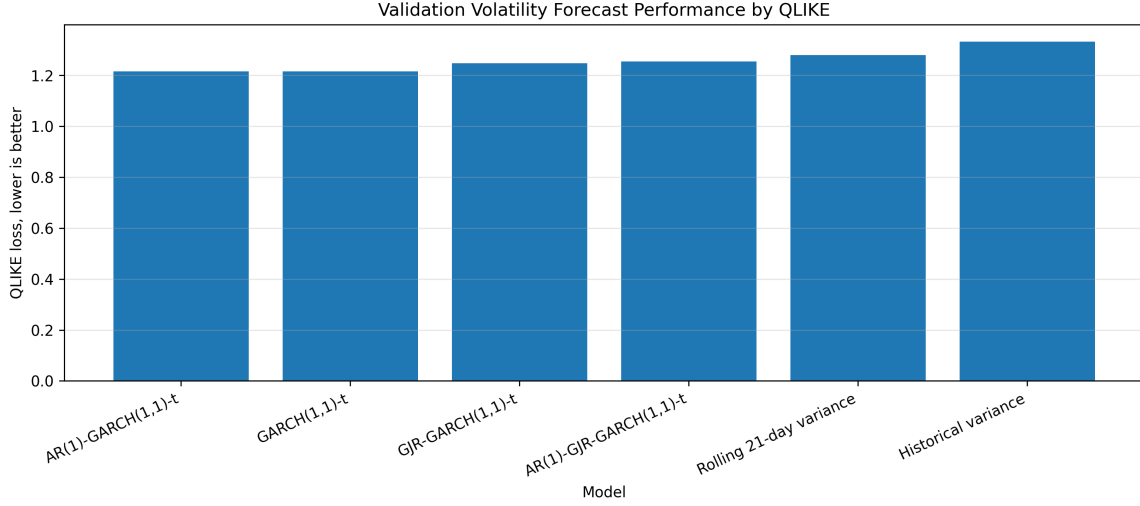


Fig. 5: Validation forecast performance by QLIKE.

TABLE IX: Test forecast performance.

Model	Obs.	MSE var.	MAE var.	QLIKE	MAE vol.	Selected
AR(1)-GARCH(1,1)-t	424	25.7583	2.0501	1.4123	0.7649	Yes
Historical variance	424	27.5620	2.0438	1.5693	0.8095	No
Rolling 21-day variance	424	29.1010	2.2409	1.7618	0.7976	No

Note. Lower values indicate better forecast performance. QLIKE is the primary criterion for volatility forecast comparison.

large shocks, the GARCH forecast increases and then decays gradually, reflecting volatility persistence. The historical variance benchmark is too flat to capture changing volatility regimes, while the rolling 21-day benchmark reacts but can remain elevated after shock episodes.

VI. DISCUSSION

The empirical results support a nuanced conclusion. Asymmetric GARCH is important for explaining VN30 volatility dynamics in-sample: the positive and significant GJR term indicates that negative shocks have a stronger effect on future volatility than positive shocks of the same magnitude. The high persistence estimate implies that volatility shocks decay slowly, which is consistent with the volatility clusters observed in the rolling-volatility plot and the conditional-volatility path.

However, the forecasting evidence is more cautious. The asymmetric AR(1)-GJR-GARCH(1,1)-t model

is not selected by validation QLIKE. Instead, AR(1)-GARCH(1,1)-t performs marginally best, with GARCH(1,1)-t nearly tied. This suggests that the additional asymmetric term improves historical interpretation but may not add enough incremental predictive information for one-step-ahead daily forecasts in the validation period. This is consistent with the broader volatility-forecasting literature, where model rankings often depend on the loss function, forecast horizon, and evaluation sample.

From a practical risk-monitoring perspective, the final GARCH forecast is still useful. It outperforms historical and rolling variance benchmarks on the test sample by QLIKE and provides a dynamic volatility signal that responds to recent shocks. Such a signal can support risk warnings, stress monitoring, and volatility-aware decision-making. At the same time, it should not be interpreted as

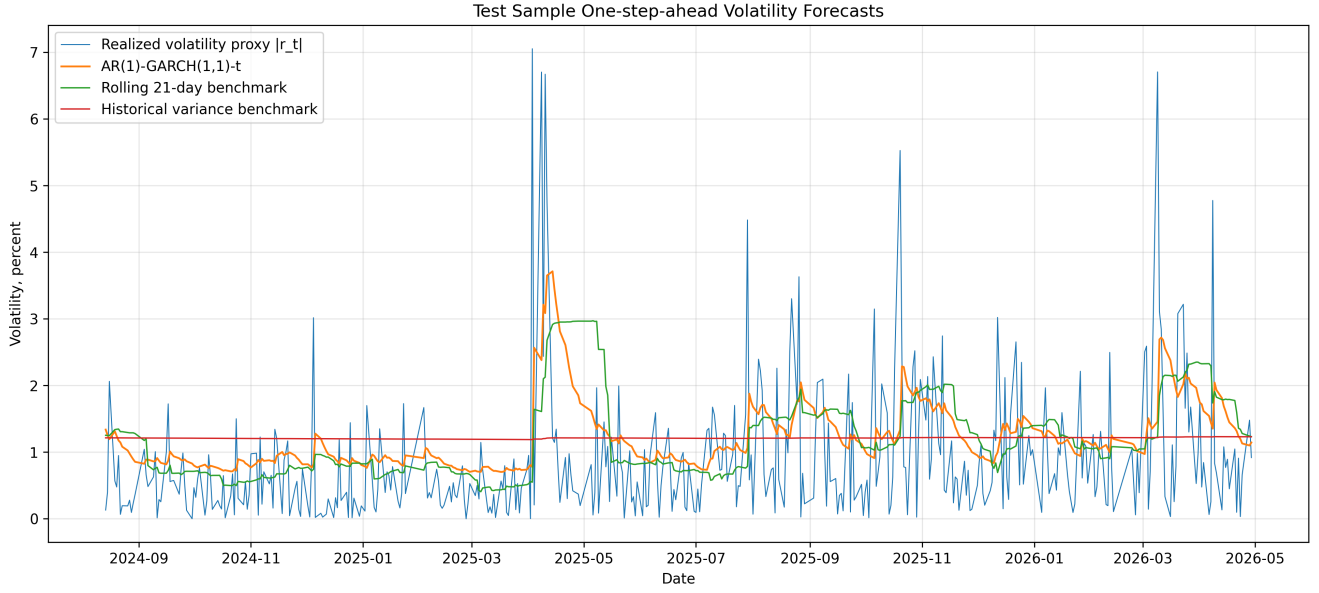


Fig. 6: Test-sample one-step-ahead volatility forecasts.

a tool for predicting the exact size of daily market jumps, because daily squared returns are noisy proxies for latent volatility.

VII. LIMITATIONS AND FUTURE WORK

Several limitations should be noted. First, the study uses squared daily returns as a volatility proxy. This is standard when realized volatility is unobserved, but it is noisy. High-frequency realized volatility would provide a cleaner evaluation target. Second, the report focuses on one-step-ahead daily volatility. Multi-step horizons may produce different conclusions about the value of asymmetric GARCH terms. Third, the validation ranking between AR(1)-GARCH(1,1)-t and GARCH(1,1)-t is extremely close. A Diebold-Mariano test or bootstrap comparison would help assess whether the difference is statistically meaningful. Fourth, the forecasting pipeline uses recursive state updating with fixed parameters within each evaluation period. Rolling-window re-estimation could be added to study parameter instability across regimes. Finally, the model is univariate and does not include macroeconomic variables, market liquidity indicators, foreign market shocks, or sector-level spillovers.

VIII. CONCLUSION

This report investigates whether asymmetric GARCH models improve the modeling and forecasting of VN30 daily volatility. The data strongly support GARCH-family modeling: VN30 returns are stationary, squared and absolute returns exhibit persistent autocorrelation, and the ARCH-LM test rejects the absence of conditional heteroskedasticity. On the training sample, AR(1)-GJR-GARCH(1,1)-t provides the best in-sample specification by BIC and residual diagnostics. Its positive asymmetric

term, high persistence, and Student-t degrees-of-freedom estimate point to three important volatility features: persistence, fat tails, and stronger responses to negative shocks.

The out-of-sample evidence leads to a more restrained forecasting conclusion. Validation QLIKE selects AR(1)-GARCH(1,1)-t, not AR(1)-GJR-GARCH(1,1)-t, as the final forecasting model. The advantage over GARCH(1,1)-t is very small, so the result should be viewed as evidence that symmetric t-GARCH models forecast at least as well as the asymmetric alternatives in this validation window. On the test sample, the final model outperforms historical variance and rolling 21-day variance benchmarks by QLIKE and most other metrics. Overall, asymmetric volatility is important for interpreting VN30 risk dynamics, but its incremental value for one-step-ahead forecasting is limited in the current empirical design.

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